Senior High School

DEPARTMENT OF EDUCATION

Learning Activity Sheet for General Mathematics

Quarter 1

LAS

2

Learning Activity Sheet for Mathematics Grade 11
Quarter 1: Unit 2

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LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	**Quarter:**	1
Lesson No.:	2.1	**Date:**	
Lesson Title/ Topic:	Sequences		
Name:		**Grade & Section:**	

I. **Activity No. 1:**

II. **Objective(s):** At the end of the activity, you should be able to determine the next term of a given sequence and apply the rule of the given sequence in finding the next term in a sequence

III. **Materials Needed:** pen and paper

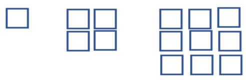
IV. **Instructions:** Analyze and solve each problem.

Activity 1: What's the Next Term?

Do the following:

a. Give the next term of the given sequences.

b. Identify the rule

1. 81, 27, 9, 3, ____
2. 
3. 7, 14, 28, 56, ____
4. 10, 21, 31, ____
5. 
6. ½, ¼, ⅛, ___
7. 8, 27, 64, __
8. 3, 6, 10, ____
9. 1x2, 2x3, 2x 4, ____
10. 5, 10, 15, 20, ____

Activity 2: Saving Money

Solve the problem and show your solution.

Scenario: A student decides to save P50 in the first week, and then increases the amount saved by P15 each week.

- Week 1: P 50
- Week 2: P 65
- Week 3: P 80
- …

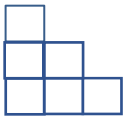
Questions:

1. How much will the student save in Week 10?
   ____
2. How much will the student have saved in total after 10 weeks?
   ____
3. If the goal is P 1500, in how many weeks will it be reached?
   ____

Activity 3: Floor Design

Solve the following problems and show your solution.

Scenario: A designer is creating a staircase using tiles based on a given pattern.



Questions:

1. What is the rule for each new step?
   ____
2. How many tiles are needed for step 10?
   ____
3. What is the total number of tiles needed for the whole staircase with 10 steps?
   ____

Activity 4 : Let's Plant Trees

Solve the problem and show your solution.

Scenario: Jean is helping organize a community project to plant trees and save the environment. She follows the pattern per group:

Questions:

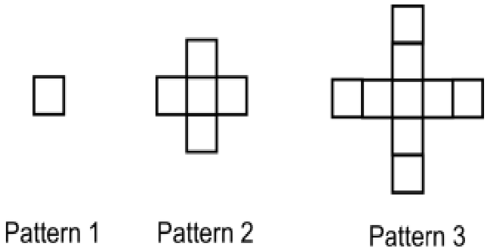
1. What is the pattern or rule in the number of trees per group?
   ____
2. How many trees will be planted in group 8?
   ____

3. What is the total number of trees planted from group 1 to 8?

____

Activity 5 : Draw Me

Given the pattern below:



Do this:

1. Draw patterns 4 and 5.

2. Make a table showing the Patterns and the number of tiles in each pattern

.

3. Identify the rule for generating the number of tiles in succeeding patterns.

4. Give a formula for the number of tiles in Pattern n.

V. Synthesis:

1. How did you arrive at your answers?

____
____
____

2. How did you find the activity? Can you share and describe your experience?

____
____
____

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	**Quarter:**	1
Lesson No.:	2.2	**Date:**	
Lesson Title/ Topic:	Sequences and Series		
Name:		**Grade & Section:**	

VI. **Activity No. :**

VII. **Objective(s):** At the end of the activity, you should be able to determine the next term of a given sequence and apply the rule of the given sequence in finding the next term in a sequence

VIII. **Materials Needed:** pen and paper

IX. **Instructions:** Analyze and solve each problem.

Activity 1: Sum It

Solve the following and show your solution.

1. Find the sum of the first 4 terms of the series:

   3 + 7 + 11 + 15

2. Find the sum of the series:

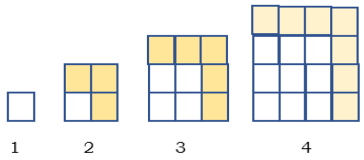
$$\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}$$

3. Write the series, 2 + 4 + 6 + 8 + 10 in sigma notation.

4. Evaluate the series written in sigma notation:

$$\sum_{n=1}^{4}(3n+1)$$

5. The figure shows a growing square-stacked of tiles. Count the total tiles in each figure.
   a. How many tiles are there in the 5^{th} stacked of tiles?
   b. How many tiles are there in total from the 1^{st} to the 7^{th} stacked tile figure?
   c. What is the rule or pattern used to determine the number of tile in each stack?



Activity 2: Saving for a Goal

Scenario:

Anna decides to save money every week to buy a bicycle worth ₱3,000. She starts by saving ₱100 in the first week and increases her savings by ₱50 each week.

Guide Questions:

1. How much will Anna save in the 5^{th} week?

2. What is the total amount saved after 5 weeks?

3. Will she reach her goal in 5 weeks? If not, how many weeks will it take?

4. How much will Anna save if she continue saving for 20 weeks? (You may use Excell spreadsheet in finding the sum)

Activity 3: SS in Disguise

Instructions for the Learners:

1. Study the following items.

   2, 4, 6, 8, 10

   1, 4, 9, 16, 25

   2 + 4 + 6 + 8 + 10

   1+ 4+ 9+ 16 + 25

2. Determine if it is a sequence or a series.

3. If it is a series, express it in sigma notation. If it is in sigma notation, translate it into its expanded series form.

4. Discuss the difference you observed between sequence and series as you complete the task.

X. Synthesis:

3. How did you arrive at your answers?

____

____

____

4. How did you find the activity? Can you share and describe your experience?

____

____

____

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	**Quarter:**	1
Lesson No.:	2.3	**Date:**	
Lesson Title/ Topic:	Arithmetic Sequence and Series		
Name:		**Grade & Section:**	

I. **Activity:** Exploring the Arithmetic Sequence and Series

II. **Objective(s):** At the end of the activity, you should be able to illustrate sequence and series. (Recommended Time: 10 minutes)

III. **Materials Needed:** Pen and Paper

IV. **Instructions:**

1. Explore how to find the sum of an arithmetic series and derive the general formula using the strategy developed by Carl Friedrich Gauss.

2. Below are two number patterns. Follow the steps carefully to discover a strategy for finding the sum of an arithmetic series.

Problem: **Find the sum of the number pattern 1, 2, 3, 4, 5, 6, 7, 8, 9, 10**

Step 1: Write the sum of the numbers in increasing order. Below it, write the sum in decreasing order.
Step 2: Pair corresponding terms in the sums you wrote in Step 1. How many pairs did you create? _______

Step 3: What is the sum of each pair? _______

Step 4: How do you get the sum of all the sums?

Step 5: Is the sum you got in step 4 the same with the sum from 1 to 10? ___________

Step 5: Since you added the series twice, divide the result by 2. ________

Step 6: Suppose you are adding numbers from 1 to 100, do the same steps above to find the sum.

Step 7. Provide a general formula for the sum of the numbers from 1 to n.

LEARNING ACTIVITY SHEET

Learning Area:	General Mathematics	**Quarter:**	1
Lesson No.:	2.4	**Date:**	
Lesson Title/ Topic:	Geometric Sequence and Series		
Name:		**Grade & Section:**	

I. **Activity:** Role Playing_Saving for a Family Trip

II. **Objective(s):** At the end of the activity, you should be able to apply knowledge of arithmetic sequence and series in real-life situation.

(Recommended Time: 30 minutes)

III. **Materials Needed:** Meta Strip

IV. **Instructions:**

Context: Imagine you and your family are planning a trip to Tagaytay. To save for the trip, you plan to save a certain amount each week, starting with ₱200 on the first week and increasing the amount by ₱50 each week.

Your task is to determine how much money you will have saved by the end of 8 weeks and how much the total savings will be after that time.

1. Understand the Problem:

2. Determine Key Information:
   - First term a_1
   - Common difference d
   - Number of terms

3. Find the amount saved in the 8th week (the last week).

4. Find the Total Savings.

5. Answer the Questions:
   - How much will you have saved in the 8th week?

   - How much total will you have saved after 8 weeks?

6. Reflection:
   - Discuss how you can apply this concept to other savings plans, like saving for a school project or a new gadget.

   - What could be the benefits of using an arithmetic sequence in planning savings?